

Frustratingly Simple Contrastive Prompt Tuning for Vision-Language Models

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Abstract

Prompt tuning, which focuses on learning continuous text prompts for adapting large vision-language models, has attracted much attention in recent years. While significant progress has been made, the growing complexity of network designs, learning algorithms, and high parameter count limits their applicability to many applications. In this paper, we introduce Contrastive Prompt Tuning (CPT) for vision-language models that simply optimizes for the learned prompts to be consistent with the image space, without any additional use of parameters or networks. In particular, CPT helps learning prompts so that the model has consistent predictions across different views of an image while also maintaining the consistency of pairwise similarities among different images. Despite being incredibly simple and easy to implement, CPT offers surprisingly good performance on a battery of datasets, outperforming existing methods for a wide variety of vision-language models.

1. Introduction

“Simplicity is the ultimate sophistication.”

Leonardo da Vinci

Large vision-language models (VLMs) [21, 25, 28, 35, 44], with appropriately designed text prompts have achieved promising progress on several downstream recognition tasks. For instance, one can prepend a category name with a prompt “a photo of a” (e.g., “a photo of a cat”) and then use as input to the CLIP [35] text encoder to classify images. However, identifying the right hand-crafted prompt is a non-trivial task, which often requires significant amount of time and domain-specific heuristics.

This has motivated much work on prompt tuning [29, 52, 53], which aims to learn soft prompts using few labeled data from the downstream tasks, while keeping the pretrained model parameters fixed. Although ubiquitous in finding better prompts compared to hand-crafted ones, the prompts learned using such methods often have poor generalization to different natural distribution shifts. To alleviate this, recent works focus on utilizing additional meta-networks [52] or prompt distribution learning [53], which are unfortu-

nately complex and increases the number of tunable parameters, defeating the whole purpose of prompt tuning for parameter efficient adaptation of large vision-language models to several downstream tasks.

In this work, our goal is to *develop a simple prompt tuning baseline for vision-language models*, one that is easy to implement yet highly effective for a wide variety of pre-trained models. One simplest approach we can think of is to augment standard prompt tuning objective with additional self-supervision without requiring any sophisticated meta-network or training strategy. In particular, we hypothesize that self-supervised learning techniques could dramatically benefit prompt tuning from a small amount of labeled examples. While such an approach looks intuitive and looks handy at first glance, it is not immediately clear whether and how contrastive learning could be exploited for prompt learning in vision-language models.

To this end, we introduce Contrastive Prompt Tuning (CPT), a “frustratingly simple” approach that explicitly optimizes for the learned prompts to be consistent in the image space. Specifically, given a few labeled examples, we augment the standard cross-entropy loss with two additional contrastive loss terms motivated by the fact that contrastive losses can improve generalization by making the model outputs invariant to small input perturbations [20, 40, 49]. The first term helps learning prompts by encouraging the model to have consistent predictions across different views of an image while the second term maintains the consistency of pairwise similarities among different images.

One particularly nice property of our approach is that it is incredibly easy to implement compared to many recent methods (e.g., CoCoOp [52], UPL [19], ProDA [29], Tip-Adapter [48]): with only few lines of code change in PyTorch, CPT can be applied to a wide variety of vision-language models, like, CLIP [35], DeCLIP [28], FILIP [46], CLOOB [10], and CyCLIP [13]. We hope our simple approach and efforts in benchmarking results of different pre-trained models (in total 10 models as opposed to prior works that only use CLIP) will open up avenues for future research in prompt learning for VLMs. We will make all our codes, data and models publicly available upon acceptance.

We evaluate CPT in four different image classification

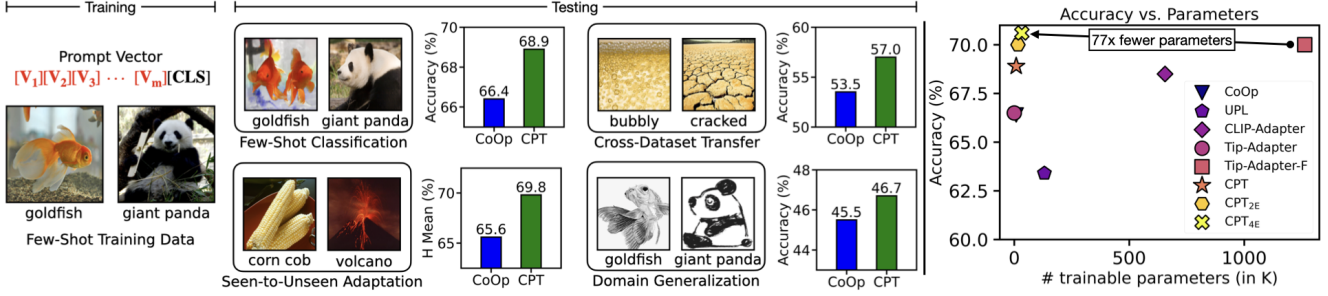


Figure 1. **Prompt Tuning for Vision-Language Models.** (Left) Figure shows four testing scenarios and the corresponding bar charts comparing the average performance of CoOp [53] and **CPT** on CLIP with ResNet50 [35]. (Right) Parameter efficiency of different methods using CLIP RN-50. Our **CPT** approach, which learns the prompts to be consistent in the image space, outperforms CoOp while it is on par or better than SOTA adaptation methods, with significantly less number of trainable parameters. Best viewed in color.

settings that can occur naturally in real-world scenarios: seen-to-unseen classes adaptation within a dataset, cross-dataset transfer, domain generalization and the standard few-shot classification setting without any distribution shift, as shown in Figure 1 (left). For few-shot classification with CLIP-RN50 [35], we observe that **CPT** improves over CoOp [53] by average 2.5% on 11 downstream datasets. For the setting of seen-to-unseen classes generalization, **CPT** yields an average 4.2% improvement over CoOp, while also very competitive with the most recent method [52] that requires an additional specialized meta-network for learning prompts (17 times more trainable parameters compared to **CPT**). The gains over CoOp are as large as 3.5% and 1.2% for the cross-dataset transfer and domain generalization settings without the need for additional unlabeled data.

Figure 1 (right) shows that despite being simple, **CPT** establishes new SOTA performance for parameter efficient adaptation of CLIP on downstream classification tasks. Interestingly, **CPT** (with only two prompts for ensembling) achieves the same performance as Tip-Adapter-F [48], while using a significant 77 times fewer trainable parameters. In summary, our findings conclusively show that **CPT** improves performance of prompt tuning across most evaluations by a significant margin, an encouraging signal for the general utility of contrastive learning in the context of generalizable prompt tuning for VLMs.

2. Related Work

Vision-Language Models. Much progress has been made in developing VLMs using single-stream [5, 26, 27, 39] or dual-stream paradigms [13, 21, 25, 28, 35, 41]. Representative works like CLIP [35] and ALIGN [21] have greatly revolutionized computer vision by allowing zero-shot transfer to a variety of downstream classification tasks. A very few methods have recently attempted learning transferable features more efficiently, using additional supervision [28, 31], finer-grained interactions [46], modern Hopfield networks [10], optimal transport distillation [44], cycle consistency [13], and hierarchical feature alignment [12].

Orthogonal to developing new learning strategies or VLM architectures, our work addresses the emerging problem of efficiently adapting large pretrained vision-language models to downstream tasks.

Prompt Tuning. Prompt tuning for efficient adaptation of vision-language models has been studied from multiple perspectives [19, 53]. Inspired by prompt tuning from NLP [24, 51], CoOp [53] minimizes the prediction error using the cross-entropy loss with respect to the learnable prompt vectors. While ProDA [29] learns diverse prompts from data to handle the variance of visual representations, UPL [19] proposes an unsupervised prompt learning framework without requiring any annotations of the target dataset. A test-time prompt tuning framework that does not need any training data or annotations to optimize the prompt is also proposed in [37]. Similar in spirit, CLIP-Adapter [11] and Tip-Adapter [48] propose to adapt vision-language models by training an additional adapter network on top of the pretrained models using a small set of labeled data. While these approaches show reasonable improvements over hand-crafted prompts, they often suffer from poor generalization under different data distribution shifts. Recently, CoCoOp [52] utilizes a Meta-Net to generate image-dependent prompt vectors for improved generalization. Alternately, we propose a much simpler yet effective method which leverages contrastive losses to learn more generalizable prompts without any additional network, making it significantly more parameter efficient than CoCoOp. In addition, **CPT** makes prompt learning extremely fast and more computationally efficient than CoCoOp, which is unwieldy to train and requires very small batch sizes during training for memory constraints.

Contrastive Learning. Contrastive learning is becoming increasingly attractive for learning robust representations of both unimodal [4, 14, 15, 33] and multimodal data [2, 35, 47]. Many variants have been recently proposed that learn representations by modeling the relationship between different instances [1, 8, 43, 50]. Contrastive learning has also been used in supervised settings, where labels are used to

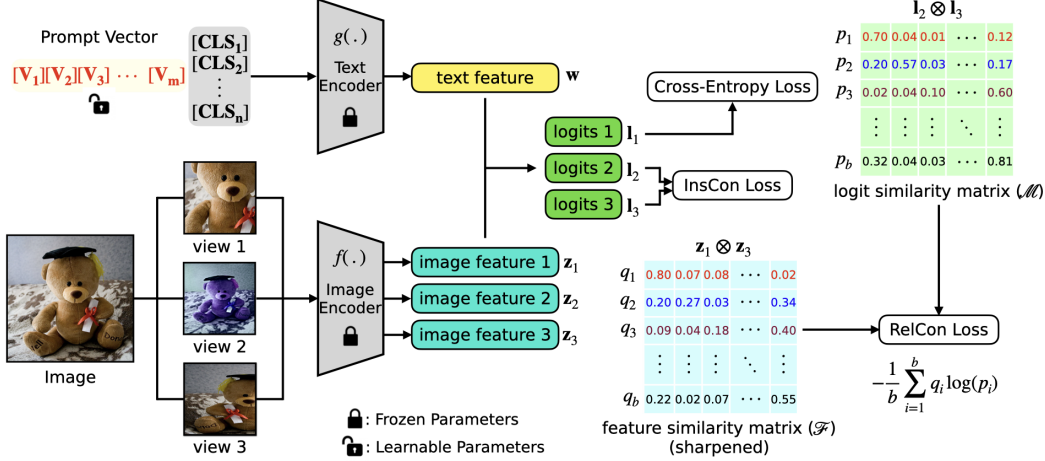


Figure 2. An overview of our Contrastive Prompt Tuning (CPT) approach. CPT learns prompt by augmenting the cross entropy loss with two self-supervised contrastive losses. The instance contrastive (InsCon) loss encourages learning instance discriminative features invariant to different views. The relational consistency (RelCon) loss makes the logit space consistent with the image space with respect to various inter-image semantic relationships. Despite being frustratingly simple, CPT is effective in learning generalizable prompts without any additional use of parameters. See Section 3 for more details. Best viewed in color.

guide the choice of positive and negative pairs [22]. While our approach is inspired by these methods, we propose contrastive prompt tuning for improving generalization in vision-language models, which to our best knowledge has not been explored in the literature.

3. Methodology

Given a pretrained vision-language model (e.g., CLIP [35]), the goal of CPT is to learn a single prompt using only a few labeled training images for efficient adaptation of the model to several downstream tasks. An overview of our approach is illustrated in Figure 2.

3.1. Preliminaries

Vision-Language Models. Dual stream VLMs jointly train an image encoder $f(\cdot)$ and a text encoder $g(\cdot)$ on data composed of image-text pairs. Given an image $\mathbf{x}$, the image encoder maps it to the feature space and outputs the l_2 -normalized image embedding $\mathbf{z} = f(\mathbf{x})/\|\mathbf{f}(\mathbf{x})\|_2 \in \mathbb{R}^d$ of dimension d . Similarly, the test description of $\mathbf{x}$ is pre-processed using an embedding layer to get $\mathbf{t}$ and is then fed to the text encoder to obtain the normalized text embedding $\mathbf{w} = g(\mathbf{t})/\|\mathbf{g}(\mathbf{t})\|_2 \in \mathbb{R}^d$. Recent VLMs (e.g. CLIP [35], DeCLIP [28], etc.) use variants of the InfoNCE loss [33] to train on large image-text data with the idea of learning perception from supervision contained in natural language.

Prompt Engineering. Once the encoders $f(\cdot)$ and $g(\cdot)$ are pretrained, using them for zero-shot prediction requires designing specific text descriptions (a.k.a prompts) to pair the test images. Given the C class names of a downstream task, generally a default prompt of “a photo of

a {class}” is used to generate the natural language class descriptions $\{\mathbf{t}_c\}_{c=1}^C$ resulting in text embeddings $\{\mathbf{w}_c\}_{c=1}^C$. For a test image $\mathbf{x}$ with embedding $\mathbf{z}$, the prediction probability is calculated as:

$$p(y|\mathbf{x}) = \frac{e^{\mathbf{z}^T \mathbf{w}_y / \tau}}{\sum_{c=1}^C e^{\mathbf{z}^T \mathbf{w}_c / \tau}} \quad (1)$$

Prompt engineering focusses on designing prompts customised to the downstream dataset to significantly improve zero-shot performance. E.g. “a photo of a {label}, a type of pet” is a more appropriate prompt for a pet classification dataset. However, prompt engineering is a manual, intuition-guided trial and error process, which can take a long time for appropriate design.

Prompt Tuning. In order to overcome the inefficiency of handcrafted prompts, prompt tuning attempts to learn continuous vectors of each token position utilizing a few labeled data. Specifically, M learnable vectors $\{\mathbf{v}_i\}_{i=1}^M$ along with the C class name word embeddings $\{\mathbf{c}_i\}_{i=1}^C$ are used to form the prompts as $\{\mathbf{t}_c\}_{c=1}^C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_M, \mathbf{c}_c\}_{c=1}^C$. The vectors $\mathbf{v}_i$ ’s can be optimized to adapt to a downstream task by propagating gradients of any loss function through the text encoder $g(\cdot)$. Till now, the use of only cross-entropy loss for prompt tuning has limited the generalization ability of the prompt to various real-world downstream tasks.

3.2. CPT: Contrastive Prompt Tuning

We propose Contrastive Prompt Tuning (CPT) which leverages self-supervised contrastive learning to learn prompts that are more generalizable to unseen classes and domains. Specifically, we achieve this by encouraging the prompt to be instance-wise discriminative while retaining

the inter-relationships between various images. As shown in Figure 2, given a few-shot dataset with C classes and the VLM encoders $\{f(\cdot), g(\cdot)\}$, our goal is to learn the M prompt vectors $\mathbf{t} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_M\}$. Given a batch of labeled images $\mathbf{x}^b = \{x_i, y_i\}_{i=1}^b$ of batchsize b , we first obtain three different views of the images $\mathbf{x}_{\text{view1}}^b$, $\mathbf{x}_{\text{view2}}^b$, and $\mathbf{x}_{\text{view3}}^b$ using a weak, strong, and weak augmentation, respectively. The images are then forwarded through the image encoder $f(\cdot)$ to obtain the corresponding image embeddings $\mathbf{z}_1, \mathbf{z}_2$, and $\mathbf{z}_3$ each of dimension $b \times d$. On the other hand, the learnable prompt vectors $\mathbf{t}$ along with the C class name word embeddings $\{\mathbf{c}_i\}_{i=1}^M$ are used to form the prompts as $\{\mathbf{t}_c\}_{c=1}^C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_M, \mathbf{c}_c\}_{c=1}^C$ and then are forwarded through the text encoder $g(\cdot)$ to obtain the text embedding $\mathbf{w}$ of dimension $C \times d$. Prediction logits are then computed as $\mathbf{l}_1 = \mathbf{z}_1 \mathbf{w}^T$, $\mathbf{l}_2 = \mathbf{z}_2 \mathbf{w}^T$, $\mathbf{l}_3 = \mathbf{z}_3 \mathbf{w}^T$ each of dimension $b \times C$. In order to capture the categorical information from the given ground truth labels, we apply a cross-entropy loss on the logit $\mathbf{l}_1$ as:

$$\mathcal{L}_{\text{CE}}(x_i, y_i) = - \sum_{k=1}^C (y_i)_k \log(\text{softmax}(\mathbf{l}_1)_k) \quad (2)$$

Use of only cross-entropy loss in few-shot setting is prone to overfitting to small training data and restricts the generalization of learned prompts to unseen images. To alleviate this, we incorporate two additional contrastive losses as follows. First, we apply an instance contrastive loss [4] on the logits $\mathbf{l}_1$ and $\mathbf{l}_2$ such that predictions for different views of the same image (positives) are similar, while that for views of different images (negatives) are different as:

$$\mathcal{L}_{\text{InsCon}}(\mathbf{l}_1, \mathbf{l}_2) = - \log \frac{\exp(\text{sim}(\mathbf{l}_1, \mathbf{l}_2)/\tau)}{\sum_{j=1}^b \mathbb{1}_{[j \neq i]} \exp(\text{sim}(\mathbf{l}_1, \mathbf{l}_j)/\tau)} \quad (3)$$

where, $\text{sim}(u, v) = u^T v / \|u\| \|v\|$ denotes cosine similarity between l_2 -normalized vectors u and v , and τ is temperature parameter. $\mathcal{L}_{\text{InsCon}}$ encourages the prompt to learn instance discriminative features from the images. Moreover, it is essential for a good prompt to capture various semantic relationships between different images to be generalizable across distribution shifts. Thus, we propose to use a relational consistency loss for prompt learning as follows:

$$\mathcal{L}_{\text{RelCon}}(\mathcal{F}, \mathcal{M}) = - \frac{1}{b} \sum_{i=1}^b q_i \log(p_i) \quad (4)$$

where, $\mathcal{F}, \mathcal{M}$ are the feature similarity matrix and the logit similarity matrix obtained as the outer products $\mathbf{z}_1 \otimes \mathbf{z}_3$ and $\mathbf{l}_2 \otimes \mathbf{l}_3$. p_i 's and q_i 's are the softmax normalized rows of $\mathcal{M}, \mathcal{F}$, with sharpening temperatures τ_1 and τ_z respectively, as shown in Figure 2. With this cross-modal design, we want to learn prompts to make the logit space consistent with various inter-image semantic relationships in the image feature space. Note that we only use the features of weakly augmented views ($\mathbf{z}_1$ and $\mathbf{z}_3$, not $\mathbf{z}_2$) to com-

pute the feature similarity matrix $\mathcal{F}$, since using aggressive augmentations can distort and hamper the capture of semantic information between different images. Finally, we iteratively optimize the total loss function $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda(\mathcal{L}_{\text{InsCon}} + \mathcal{L}_{\text{RelCon}})$ and update the learnable prompt through standard backpropagation. λ is a weight to balance the impact of contrastive loss terms. To reduce the number of hyper-parameters, we use the same weight λ for both $\mathcal{L}_{\text{InsCon}}$ and $\mathcal{L}_{\text{RelCon}}$ in all our experiments. Note that the pretrained vision-language model is frozen and the prompt is the only learnable parameter in our approach.

Additionally, we incorporate a memory buffer of size $100 \times$ of the batch size, to instill information from a diverse set of images and their views, which we find is essential for learning a generalizable prompt. Furthermore, as a regularizer and making learning prompts invariant to the position of the class token, we randomly choose the position of the class token between “start”, “mid”, and “end” in every iteration following an uniform distribution. Algorithm 1 summarizes the proposed approach in PyTorch-style pseudocode. Once the training is completed, the learned prompt can be used with the class embeddings appended at the end for any desired downstream task.

Algorithm 1 : CPT in a PyTorch-like style.

```

1  # image_encoder f, text_encoder g
2  # I[b, h, w, c] - minibatch of images
3  # T[b, l] - minibatch of texts
4  # feature memory buffer - FMB [d, 100b]
5  # logits memory buffer - LMB [n_cls, 100b]
6  # generate views
7  I_view1 = weak_aug(I)
8  I_view2 = strong_aug(I)
9  I_view3 = weak_aug(I)
10 # extract feature representations
11 z1 = f(I_view1) #[b, d]
12 z2 = f(I_view2) #[b, d]
13 z3 = f(I_view3) #[b, d]
14 dequeue_and_enqueue(FMB, z3)
15 w = g(Prompt Vector) #[n_cls, d]
16 # obtain logits [b, n_cls]
17 l1 = (z1 @ w.T)
18 l2 = (z2 @ w.T)
19 l3 = (z3 @ w.T)
20 dequeue_and_enqueue(LMB, l3)
21 # compute cross-entropy loss
22 loss_CE = CrossEntropy(l1, labels)
23 # compute instance-wise contrastive loss
24 loss_InsCon = SimCLR(l2, l3)
25 # compute relational consistency loss
26 feat_sim_mat = softmax(z1 @ FMB.T / tau_z, dim=1) #
   [b, 100b]
27 logit_sim_mat = softmax(l2 @ LMB.T / tau_l, dim=1) #
   [b, 100b]
28 loss_RelCon = torch.sum(-feat_sim_mat * log(
   logit_sim_mat), dim=-1).mean()
29 # total loss
30 loss = loss_CE + loss_InsCon + loss_RelCon
31 # compute gradients and optimize
32 loss.backward()
33 optimizer.step()

```

4. Experiments

In this section, we examine our contrastive prompt tuning approach to answer three key research questions. Q1: Can **CPT** improve prompt tuning in few-shot classification setting without distribution shift? Q2: To what extent contrastive learning benefits generalization of prompt tun-

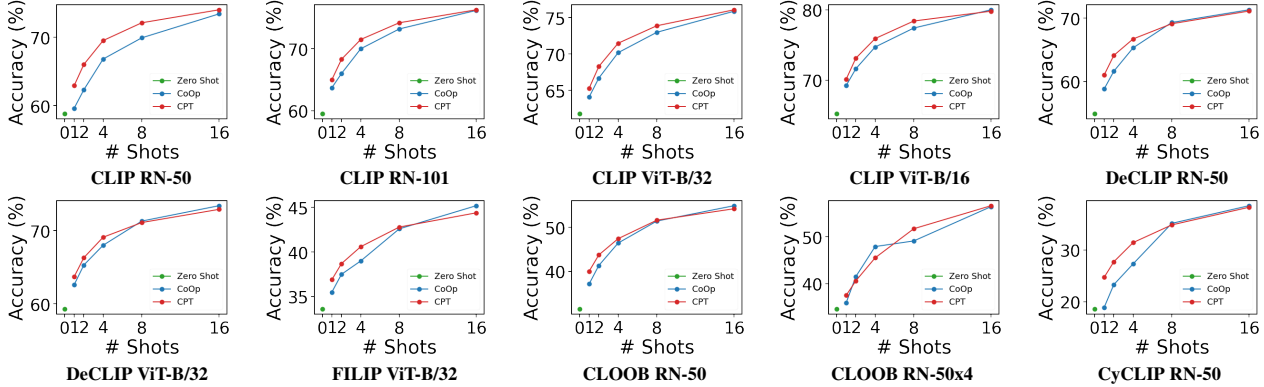


Figure 3. **Few-shot Classification.** **CPT** consistently outperforms CoOp across all ten VLMs, showing the effectiveness of contrastive prompt tuning for efficient adaptation of pretrained models. We report average accuracy across the 11 datasets for each few-shot setting.

ing when learned prompts are transferred across different classes and datasets? Q3: Can **CPT** be universally effective across a wide range of pretrained VLMs of different sizes?

4.1. Experimental Setup

Datasets. Following [52, 53], we evaluate **CPT** using 15 downstream classification datasets, including general object recognition (ImageNet [7] and Caltech101 [9]), fine-grained recognition (OxfordPets [34], StanfordCars [23], Flowers102 [32], Food101 [3] and FGVCAircraft [30]), scene recognition (SUN397 [45]), texture recognition (DTD [6]), satellite image recognition (EuroSAT [16], action recognition (UCF101 [38]), and four ImageNet variants with domain shifts (ImageNetV2 [36], ImageNet-Sketch [42], ImageNet-A [18] and ImageNet-R [17]). We use first 11 datasets for few-shot classification, seen-to-new classes adaptation, and cross-dataset transfer experiments. For domain generalization, we use ImageNet as source dataset and four of its variants as target datasets. We use the standard splits provided in [53] for training and the original test/validation set for testing on all datasets.

Models. We experiment with 10 publicly available pretrained VLMs of varying architectures and sizes from CLIP [35], DeCLIP [28], FILIP [46], CLOOB [10], and CyCLIP [13]: CLIP ResNet-50, CLIP ResNet-101, CLIP ViT-B/32, CLIP ViT-B/16, DeCLIP ResNet-50, DeCLIP ViT-B/32, FILIP ViT-B/32, CLOOB ResNet-50, CLOOB ResNet-50x4, CyCLIP ResNet-50.

Baselines. We compare our approach with the following baselines. (1) Zero-shot CLIP that uses hand-crafted prompts for downstream classification, (2) CoOp [53] that learns prompt by only minimizing the cross-entropy loss, (3) a state-of-the-art prompt tuning method for CLIP, Co-CoOp [52] that uses an additional meta-network for predicting prompts. We also compare with recent CLIP adaptation methods including CLIP-Adapter [11], and Tip-Adapter [48] in few-shot classification settings. We directly quote numbers reported in published papers when possible

or use the source code released by CoOp [53] authors under same experimental settings for a fair comparison.

Implementation Details. Following [53], we set the number of tokens in each prompt to 16 with random initialization for all the experiments except for the seen-to-unseen classes adaptation and experiments in Table 2 and Table 3, where we set it to 4 and initialize with the word embeddings of “a photo of a” as in [52]. For few-shot classification, we follow CLIP [35], which learns with 1, 2, 4, 8, and 16 labeled samples per class on each downstream task. The loss weight coefficient is set to $\lambda = 0.1$. The temperature values τ , τ_z and τ_l are set to 0.5, 0.04 and 0.07, respectively. We use a batch size of either 8 or 32, except in ImageNet for which we used a batch size of 128, and train for either 50 or 200 epochs based on the dataset with a learning rate of 0.002. For generating multiple views, we compose strong augmentations using RandAug, color jittering, random grayscaling and blurring, while for weak augmentation we simply use random resized cropping and random horizontal flipping. For ensembling, we independently train the prompts with different initialization, and average the logits from all the prompts during inference. We run all experiments for three times with different random seeds and report the mean numbers in all our testing scenarios. We use one NVIDIA Tesla A100 GPU for learning all our prompts.

4.2. Few-Shot Classification

First, we consider the standard few-shot classification in which we study the performance on test data belonging to same classes and same domain as the training data.

Comparison with CoOp. Figure 3 shows the results on 10 different VLMs. While using CLIP RN-50 with single labeled images per class, CoOp outperforms handcrafted prompts (Zero shot) by 0.8% on average, while **CPT** provides 4.1% improvement. When compared to CoOp, the gain is particularly significant in the low-shot scenarios, which are practically important cases. E.g. for CLIP RN-50 backbone, the improvement over CoOp in 1-shot is 5.9

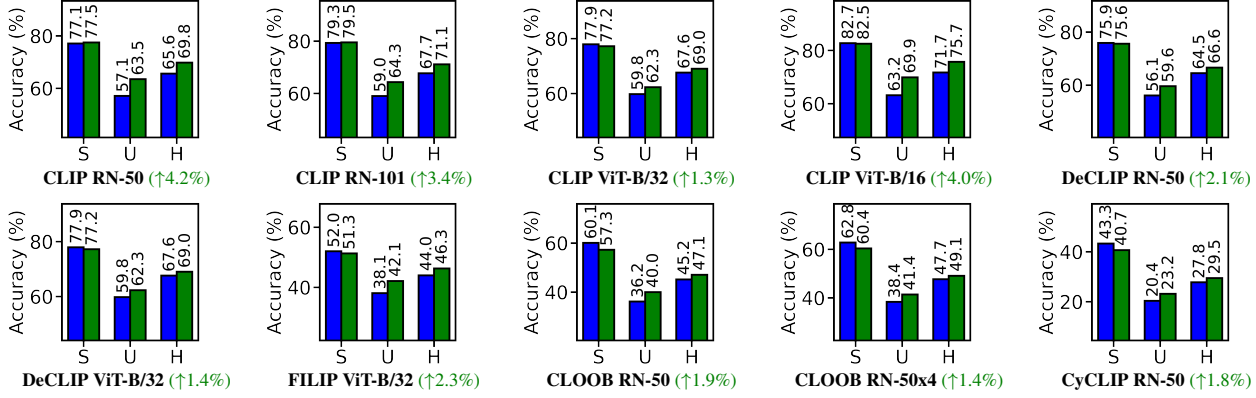


Figure 4. **Seen to Unseen Classes Adaptation.** Figure shows bar charts comparing the average performance on 11 datasets of **CPT** with CoOp on seen classes (S), unseen classes (U), and their harmonic mean (H) on 10 varieties of VLM backbones. Our **CPT** approach outperforms CoOp consistently on all the models. The blue bars represent CoOp, green bars represent **CPT**. Best viewed in color.

Table 1. **Generalization from seen to unseen classes.** We report accuracy with CLIP ViT-B/16 model on seen classes (S), unseen classes (U), and harmonic mean of both of them (H). **CPT** outperforms CoOp by +4.0% while performing at par with parameter heavy CoCoOp. To compete with CoCoOp, we adopt a 2× ensemble **CPT**_{2E} which outperforms CoCoOp despite having significantly less parameters.

Method	#Params	Average			ImageNet			Caltech101			OxfordPets			StanfordCars			Flowers102		
		S	U	H	S	U	H	S	U	H	S	U	H	S	U	H	S	U	H
ZS CLIP	–	69.3	74.2	71.7	72.4	68.1	70.2	96.8	94.0	95.4	91.2	97.3	94.1	63.4	74.9	68.7	72.1	77.8	74.8
CoOp	2.05K	82.7	63.2	71.7	76.5	67.9	71.9	98.0	89.8	93.7	93.7	95.3	94.5	78.1	60.4	68.1	97.6	59.7	74.1
CoCoOp	35.38K	80.5	71.7	75.8	76.0	70.4	73.1	98.0	93.8	95.8	95.2	97.7	96.4	70.5	73.6	72.0	94.9	71.84	81.7
CPT	2.05K	82.5	69.9	75.7	76.4	69.1	72.6	98.0	94.3	96.1	95.4	97.8	96.6	74.5	71.6	73.0	97.5	66.6	79.2
CPT _{2E}	4.10K	83.2	71.8	77.1	76.6	70.6	73.5	98.2	94.3	96.2	95.9	98.2	97.1	75.6	72.3	73.9	97.7	72.2	83.0
Method	#Params	Food101			FGVCAircraft			SUN397			DTD			EuroSAT			UCF101		
		S	U	H	S	U	H	S	U	H	S	U	H	S	U	H	S	U	H
ZS CLIP	–	90.1	91.2	90.7	27.2	36.3	31.1	69.4	75.4	72.2	53.2	59.9	56.4	56.5	64.1	60.0	70.5	77.5	73.9
CoOp	2.05K	88.3	82.3	85.2	40.4	22.3	28.8	80.6	65.9	72.5	79.4	41.2	54.2	92.2	54.7	68.7	84.7	56.1	67.5
CoCoOp	35.38K	90.7	91.3	91.0	33.4	23.7	27.7	79.7	76.9	78.3	77.0	56.0	64.9	87.5	60.0	71.2	82.3	73.5	77.6
CPT	2.05K	89.9	90.9	90.4	38.7	28.4	32.8	80.9	74.4	77.5	80.3	47.7	59.8	92.2	58.1	71.3	83.5	70.3	76.3
CPT _{2E}	4.10K	90.2	91.4	90.8	40.5	31.6	35.5	81.5	76.3	78.8	81.8	51.5	63.2	93.1	57.7	71.2	84.3	73.8	78.7

times that in 16-shot, which concretely affirms the advantage of self-supervised contrastive learning in overcoming overfitting and better generalization. Similarly, in one-shot setting of CyCLIP RN50 and DeCLIP ViT-B/32, **CPT** outperforms CoOp by +5.8% and +1.1% respectively, showing its effectiveness in few-shot classification across different models. In 2-shot and 4-shot settings, **CPT** shows 2.0% and 1.4% improvement on average over all the models.

4.3. Generalization from Seen to Unseen Classes

Following [52], we show the generalization performance of different methods, namely CoOp [53] and CoCoOp [52] including **CPT** by training on seen (base) classes while evaluating on both seen and unseen (new) classes.

Comparison with Vanilla Prompt Tuning (CoOp). We first compare our approach **CPT** with the vanilla prompt tuning, CoOp to show how much performance improvement **CPT** can achieve across different VLMs. As shown in Figure 4, **CPT** consistently outperforms CoOp in improving

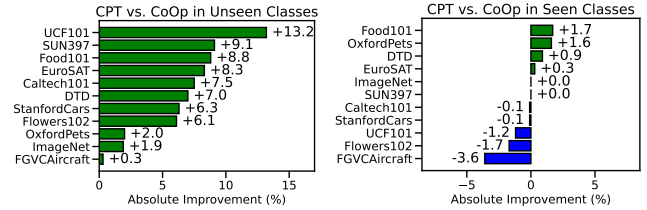


Figure 5. **Absolute improvement over CoOp w/ CLIP RN-50.** Bar charts show improvement over CoOp on seen and unseen classes for each datasets. Best viewed in color.

generalization performance across a wide variety of models. The gains over CoOp are as large as 4.2% on CLIP-RN-50, conforming the hypothesis that contrastive learning can significantly improve generalization of learned prompts to recognize unseen classes. Figure 5 shows absolute improvement over CoOp on both seen and unseen classes for each of the 11 downstream datasets. As expected, **CPT** improves the performance of CoOp in unseen classes on all datasets (see Figure 5 (left)), while performance drops marginally in

Table 2. **Cross-dataset transfer.** Prompts trained on ImageNet using **CPT** are more generalizable to other datasets than CoOp, while competent with CoCoOp. A simple $2\times$ ensemble **CPT**_{2E} fills the gap while being $8.6\times$ parameter efficient. All the baseline use CLIP ViT-B/16 backbone under the same experimental settings.

	# Params	Source											Avg
		IN1K	Caltech	Pets	Cars	Flowers	Food	Aircraft	SUN	DTD	EuroSAT	UCF	
CoOp	2.05K	71.5	93.7	89.1	64.5	68.7	85.3	18.5	64.2	41.9	46.4	66.6	63.9
CoCoOp	35.38K	71.0	94.4	90.1	65.3	71.9	86.1	22.9	67.4	45.7	45.4	68.2	65.7
CPT	2.05K	71.3	94.1	90.2	64.8	70.6	85.9	21.8	66.3	43.6	46.0	68.0	65.1
CPT _{2E}	4.10K	71.6	94.2	90.3	64.5	70.9	86.3	21.7	66.8	45.3	47.7	69.3	65.7

Table 3. **Domain generalization.** **CPT** outperforms both CoOp and CoCoOp, while using same number of tunable parameters as CoOp. All the baseline use CLIP ViT-B/16 backbone under the same experimental settings.

	# Params	Source					Target		
		ImageNet	ImageNetV2	ImageNet-Sketch	ImageNet-A	ImageNet-R			
CoOp	2.05K	71.5	64.2	48.0	49.7	75.2			
CoCoOp	35.38K	71.0	64.1	48.8	50.6	76.2			
CPT	2.05K	71.3	64.1	49.0	50.7	76.4			
CPT _{2E}	4.10K	71.6	64.6	49.2	51.1	76.8			

the base classes of few datasets (see Figure 5 (right)).

Comparison with CoCoOp. Table 1 shows the comparison of our **CPT** approach with CoCoOp including Zero-shot CLIP (ZS CLIP) and CoOp under the same experimental settings (w/ CLIP ViT-B/16). As expected, **CPT** significantly outperforms ZS CLIP on all datasets as hand-crafted prompts are naturally worse in generalization, while learnable prompts has the ability to learn the intricate differences between the finely differing categories from data. When compared to CoCoOp, **CPT** achieves very competitive average performance without requiring any additional meta-network as in CoCoOp [52]. However, by simply ensembling two learned prompts, **CPT**_{2E} can outperform CoCoOp on majority of the datasets, to obtain the best average performance of 77.1% across all datasets. This is especially significant as our approach achieves greater performance at the cost of significantly less number of trainable parameters compared to CoCoOp ($8.6\times$ lower).

4.4. Cross-Dataset Transfer

In this section, we show CPT’s ability to transfer learned prompt beyond a single dataset. This is fundamentally more challenging compared to generalizing well while remaining within the same data distribution. In this setting, we train using the generic and natural image dataset ImageNet and test the efficacy of the learned prompt in 10 different datasets comprising of images coming from finegrained categories like Cars, Flowers, Food, Aircraft etc or texture classification like DTD. As seen from Figure 6 (left), while using CLIP RN-50 as the backbone, **CPT** demonstrates better transferability than CoOp on all the datasets, leading to an average accuracy of 57.0%, which is $+3.5\%$ better than CoOp. Likewise for CLIP ViT-B/16, Table 2 shows that per-

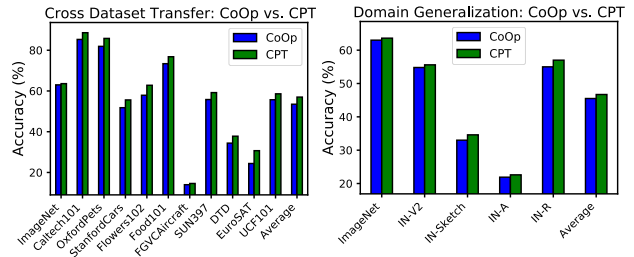


Figure 6. (Left) Cross Dataset Transfer and (Right) Domain Generalization using CLIP RN-50.

formance of **CPT** is comparatively better than CoOp which uses same number of learnable parameters. We also achieve similar performance to CoCoOp while being $8.6\times$ parameter efficient with $2\times$ ensemble **CPT** and $17.2\times$ parameter efficient with **CPT**. In summary, these results show that a contrastively learned prompt not only transfers the knowledge to very different settings but also does it in much more parameter efficient manner.

4.5. Domain Generalization

Domain or distribution shifts are very common in the real-world. In order to study the robustness of the learned prompts to out-of-distribution data, following [52], we learned a prompt on the ImageNet dataset and tested its performance on 4 of its specially designed benchmarks possessing distribution shift, like ImageNetV2, ImageNet-Sketch, etc. Table 3 clearly shows the dominating performance of **CPT** over CoOp and CoCoOp even with a single prompt for CLIP ViT-B/16. Using an additional prompt to ensemble even pushes the performance further by 0.38% on average over the target datasets, while still using 8.6 times less parameters than CoCoOp. Likewise, similar trends are

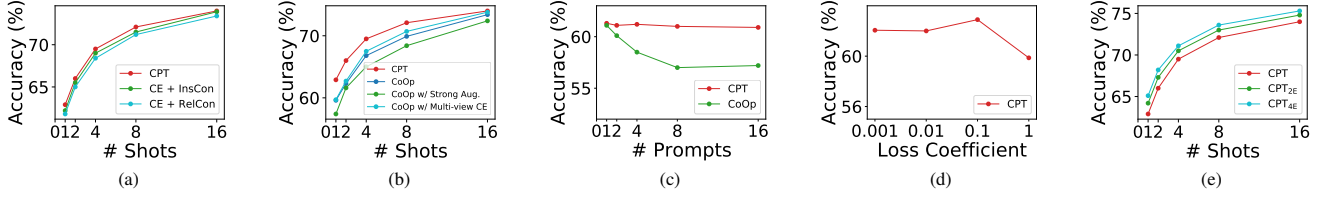


Figure 7. **Ablation Studies.** All results are based on CLIP RN-50 model. (a) **Effect of losses.** studies the effectiveness of different losses. (b) **CoOp with strong augmentations.** shows average downstream performance using additional strong augmentations in CoOp. (c) **Number of learnable prompt tokens.** shows variation of accuracy with the number of learnable tokens for 1-shot experiment on ImageNet. (d) **Effect of loss coefficient.** studies the effect of different values of the hyperparameter λ in 1-shot on all datasets. (e) **Effect of Ensembling.** shows the average performance improvements with ensembling. Best viewed in color.

observed for CLIP RN-50 as shown in Figure 6 (right). This highlights the effectiveness of **CPT** in learning domain-invariant prompts while being highly parameter efficient.

4.6. Ablation Studies

Effect of Losses. To study the effectiveness of both contrastive losses, we obtain the few-shot classification performance by using either of the losses $\mathcal{L}_{\text{InsCon}}$ and $\mathcal{L}_{\text{RelCon}}$, independently with $\mathcal{L}_{\text{CE}}$. Figure 7(a) shows the average accuracy on 11 datasets for CLIP RN-50. Only using $\mathcal{L}_{\text{InsCon}}$ gives an average improvement of 2.0% over CoOp, while only using $\mathcal{L}_{\text{RelCon}}$ provides 1.6% average improvement. The best performance is obtained with both of the losses, with an average improvement of 2.5%, showing the effectiveness of both instance discrimination and relational consistency in learning effective generalizable prompts.

CoOp with Strong Augmentations. CoOp [53] does not use any strong data augmentations. So, *is the obtained improvement in CPT just because of additional data augmentation?* To answer this, in Figure 7(b) we run CoOp with strong augmentation as well as using augmentations to generate multiple-views of the same image and using a cross-entropy loss to train on them on CLIP RN-50 for all the 11 datasets. Interestingly, using only strong augmentations in CoOp reduced the performance by 1.4% on average, while applying cross-entropy loss on multiple-views improved the average performance by 0.5%, which still lags behind **CPT** by 2.0%. This corroborates the fact that our improved performance is not due to data augmentation instead due to the proposed contrastive losses for avoiding overfitting and under-performance of prompt learning in few-shot settings.

Number of Learnable Prompt Tokens. Figure 7(c) shows the effect of number of learned prompt tokens on few-shot classification performance for 1-shot setting on ImageNet dataset. An interesting observation is that the performance for CoOp increases with the reduction in the number of prompt tokens (an increase of 3.9% from 16 to 1 tokens). This highlights the problem of potential overfitting in the low-shot setting and how CoOp is prone to it. On the other hand, **CPT** maintains a very stable performance (a drop of only 0.4% from 1 to 16 tokens) across the number of tokens, demonstrating the importance of contrastive learning

in learning effective prompts in low-shot settings.

Effect of Hyperparameters. Figure 7(d) shows the effect of loss coefficient λ , where we vary λ with values 0.001, 0.01, 0.1, 1.0 for 1-shot setting on all the downstream tasks. We find $\lambda = 0.1$ to be the best value on average and use it for all experiments. Study of performance by varying τ_1 can be found in the supplementary material.

Effect of Ensembling. In Figure 7(e) instead of learning a single prompt, we learn an ensemble of two (**CPT**_{2E}) and four (**CPT**_{4E}) prompts on CLIP RN-50. Simply learning two prompts independently gives an average improvement of 1.1%, while learning four prompts provides 1.7% improvement. The performance improvement with ensembles is owed to the increase in learnable parameters and diverse knowledge through independent training which has been also studied in prompt learning [35, 53].

Initialization of Prompts. We initialize the prompts with word embeddings of a handcrafted prompt before training, and saw no major changes in the performance. E.g. learning 4 tokens for ImageNet 1-shot using “a photo of a” as initialization yields 61.1% compared to 61.2% using random initialization, in consistent with the findings in [53].

Additional experimental results and discussions are included in the supplementary material.

5. Conclusion

In this paper, we propose a simple, efficient and effective prompt tuning approach for vision-language models. Specifically, we augment the standard cross-entropy loss with two additional contrastive losses that optimizes for the learned prompts to be consistent with the image space. We demonstrate the effectiveness of our approach on multiple diverse datasets, outperforming state-of-the-art methods, without any additional use of parameters. One current limitation of our work is it only focuses on object/image recognition. However, **CPT** is so simple that with minor modifications, it can be applied to other important vision tasks like object detection, semantic segmentation, and action recognition, which will be studied in future work.

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